Current Trends in Software Quality

Presented To

Dr. Hasan Sarwar

Professor

Department of Computer Science and Engineering

United International University

Presented By

Nusrat Jahan Sarna	0122410015
Farzana Ahmed Rithen	0122410051
Md. Nazmuz Sakib	0122410053
Sumaiya Akter	0122320002
Rayhan Al Shorif	0122410022

Table of contents

#1 Introduction

#2 Research Methodology

#3 Challenges in Implementing Software Quality Practices

#4 Conclusion

#5 References

Introduction

What is Software Quality?

Software quality is the measure of how well a software product meets user requirements, expectations, and standards, encompassing aspects such as functionality, reliability, usability, efficiency, maintainability, and portability.

Objectives of the Analysis

Find out the current trend in software quality.

Importance of Software Quality in the Industry

The importance of software quality in the industry lies in ensuring reliable, efficient, and user-friendly software.

Research Methodology

Trending Works

Title	Year	Main Focus/ Contribution	Limitations/Gap
ROLE OF ARTIFICIAL INTELLIGENCE (AI) TOOLS FOR ASSURING QUALITY IN SOFTWARE [1]	2022	The paper explores the impact of AI on Software Quality Assurance (SQA), focusing on validation, functionality, methodology, and potential strategies in software quality testing. The main objectives are to identify the effects of implementing AI tools in SQA and to address the challenges faced when using AI in software testing. The study examines four AI tools: Testim and Test Craft (web applications), and Applitools and Functionize (web and mobile applications). The research follows three phases:	● AI strategies are often very labor intensive. The worldwide cost-benefit ratio can be too high when the strategies are implemented incorrectly or when there are negligible marginal effects prior to using more sophisticated methods. ● Insufficient budget. ● Defects and malfunctioning. ● Limited knowledge of techniques and tools.

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
Cont..[1]		1. Data Collection: Qualitative methods, including questionnaires and interviews, are used to gather data, with responses recorded on a scale from 1 to 5. 2. Scrutiny: The data is scrutinized to ensure the maximum number of accurate and reliable responses for final analysis. 3. Statistical Analysis: Qualified responses are analyzed using statistical tools to ensure the results meet the study's objectives. The survey results indicate that out of 10 software houses in Pakistan, only 2 use AI tools for software quality assurance, highlighting a significantly low adoption rate.	

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
Review Dynamics and Their Impact on Software Quality [2]	2020	● Investigates code review dynamics. ● Examines impact of visible information (comments and votes) on reviewer decisions. ● Explores relationship between review dynamics and defect-proneness of patches.	● Specific Case Studies ● Focus on Review Dynamics Limited Scope of Metrics ● Relatively Small Impact on Software Quality

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
A framework for software quality measurement [3]	2019	This paper represents a comprehensive framework for software quality measurement, integrating multiple quality attributes from accepted guidelines like ISO. The framework combines automated tools, user input, and static and dynamic analysis, which will help create software systems that are more dependable, effective, and easy to use.	There are different software systems or organizations that follow different standard and measurement techniques and software needs here in this paper they are not considered. There is not enough reliable data available to validated the quality measures

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
Software Engineering: A Quality Management Perspective [4]	2007	i) This study aims to investigate the notion of quality in the context of software development by utilizing the ISO, TickIT and CMM frameworks. ii) This study also aims to emphasize how the various viewpoints of persons engaged in software development would affect the perception and assessment of quality. iii) The approach & methodology adopted for this paper were a review of the literature and best practice in software engineering.	i) Research Limitation: In light of software engineering practice, this study contributes to the definition of the advantages and disadvantages of ISO, TickIT and CMM. ii) Practical Limitations: It is demonstrated in this paper that software engineers should be more proactive in addressing quality concerns rather than being reactive when it comes to performance and conformance difficulties in software projects

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
The Evolving Landscape of Software Quality Assurance: Challenges and Opportunities[5]	2023	1. Software Quality Assurance (SQA) is changing and evolving, focusing on the new challenges and opportunities that have emerged. 2. Highlights the vital role of SQA in ensuring software works well and reliably, especially as technology advances quickly. 3. Provides advice for SQA professionals and developers on how to handle current challenges and take advantage of new opportunities.	• Faster development cycles. • Ensuring software works across different platforms.

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
Cont.[5]	2023	4. Highlights the benefits of combining traditional methods with new technologies like Agile, DevOps, AI, and automation to enhance SQA practices.	• Managing the complexity of interconnected software systems.

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
Developing Software Quality and Testing Capabilities in Hispanic America: Challenges and Prospects [6]	2020	1. The paper examines the evolution of computing in Latin America from the late 1950s, highlighting the influence of the USA and Western Europe. 2. It discusses the growth of computing-related education and IT ecosystems and the variations in IT development among different countries.	• Building scientific and technological research and development capabilities remains a challenge • Smaller countries have lagged in adopting computing technologies compared to larger nations.

Trending Works (Cont.)

Title	Year	Main Focus/ Contribution	Limitations/Gap
Cont.[6]	2020	3. The role of industrial IT players in providing training and the typical 1-5 year lag in IT transitions are also covered. 4. The paper emphasizes the research and development efforts in Brazil and Chile, the impact of the Internet and mobile communications, and the rising demand for software due to Digital Transformation.	• Ensuring quality and productivity in the software industry, updating curricula, building competencies, and adapting to rapid digital changes are ongoing issues, especially in meeting the demands of Digital Transformation.

Challenges in Implementing Software Quality Practices

Technical Challenges

Organizational Challenges

Cost and Resources Challenges

Conclusion

- Software Quality ensures the delivery of :
  - high-quality
  - reliable
  - user-friendly software products

- By adopting modern trends organizations can significantly enhance their software quality practices.

- A good quality software meet user needs, cost effective , build trust, and achieve long-term success

- Analyzed from papers study we understand new trends of software quality and their current works

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Thank You!